

FDMA, so if one frequency band had, say, three time-slots, it automatically multiplied the capacity of the network thrice.

m Code Division Multiple Access (CDMA)

In the CDMA system, calls were separated by a unique code assigned to each of them.

FDMA, TDMA and CDMA are explained in detail in the next chapter.

Till the 1980s, there was no standardisation for cellular phone systems despite constant evolving technology in the field, causing much confusion and some dismay with respect to compatibility with other mobile phone networks. Standardisation was essential if people were to roam the world and still be able to connect to a telephone network.

In 1982, the GSM (*Groupe Spécial Mobile*) group was founded to address these issues. Three years were spent mulling over whether the global standard should be analogue or digital till finally, in 1985, after much discussion and many trial runs, they decided that the world should use digital mobile telephony. In 1987, they chose TDMA as their solution and by 1990, the first GSM Standard was published. To keep the acronym 'GSM' alive, their standard was called the Global System for Mobile Communication.

The GSM Standard introduced the SIM (*Subscriber Identity Module*) card, which held information about the user and provided memory to store phone numbers and text messages. The SIM card could be shifted from handset to handset, allowing users to choose handsets according to their fancy without having to bother about their cellular service provider. It is estimated that approximately 68 per cent of the world's cellular phone subscribers today are on a GSM network.

By 1993, Qualcomm had proposed a standard called cdmaOne, based on the CDMA technique. Unlike FDMA or TDMA, CDMA could theoretically handle an obscenely large number of callers.